# Multi-Horizon Stock Volatility Forecasting

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**Abstract.** We evaluate whether a deep sequence model or a cross-sectional graph-attention branch changes multi-horizon daily Parkinson-variance forecasts relative to a linear Heterogeneous Autoregressive (HAR) baseline on the Vietnamese equity market — VN30 and VN100 — under panel-correct inference. Three models share the same five node features: a linear HAR, a five-feature LSTM, and a five-feature LSTM+GAT (directed volume→Parkinson Top-5 weighted two-hop attention). We report five metrics (MSE, RMSE, MAE, QLIKE, $R^2$) with equal weight and assess significance with the date-clustered Diebold–Mariano (DM) test; predictions are floored at $10^{-2}$ times the per-node training mean. The primary target is short-term (horizons $h \in \{1, 5\}$); $h \in \{10, 22\}$ are an extended study. HAR has the lowest QLIKE on both panels at every horizon, and no learned model has a significantly lower QLIKE than HAR (date-clustered DM, $p \geq 0.05$). At $h1$ and $h5$ the deep models have the lowest MAE on VN100 (LSTM vs HAR, $p < 0.001$), and the LSTM+GAT has a significantly lower QLIKE than the no-graph LSTM in both panels (VN100 $h1$ $p < 0.001$, $h5$ $p = 0.022$; VN30 $h1$ $p < 0.001$, $h5$ $p = 0.031$), equal to HAR's QLIKE level at VN100 $h5$ (0.5690 vs 0.5633). At $h10$ and $h22$ HAR has the lowest MSE, RMSE, QLIKE and $R^2$, and the LSTM+GAT-vs-LSTM QLIKE difference is not significant ($p \geq 0.55$). The ranking is metric- and horizon-dependent, so all five metrics are reported with equal weight. Methodologically, a naive per-observation DM test on this cross-sectionally dependent panel overstates significance by roughly the square root of the number of stocks.

**Keywords:** Volatility forecasting · HAR · LSTM · Graph Attention Network · Diebold–Mariano.

## 1 Introduction

Daily volatility forecasting underpins risk management, position sizing and derivative pricing. The HAR model of Corsi [1] is a linear benchmark: a few OLS coefficients on daily, weekly and monthly aggregates of realized volatility capture much of the long-memory structure. Deep sequence models and graph neural networks are frequently proposed to capture nonlinear temporal dynamics and cross-sectional spillovers, and the evidence that they beat HAR out of sample is mixed; positive claims are often confounded by data-design choices, such as a common-date intersection that discards most of the panel or a naive significance test on cross-sectionally dependent data.

This paper measures, under a single panel-correct evaluation on the Vietnamese equity market, whether the temporal nonlinearity or the cross-sectional graph changes the forecast relative to a linear HAR. We ask: (1) Does a deep temporal LSTM over the shared feature set change the out-of-sample forecast relative to the linear HAR, and at which horizons and metrics? (2) Does a cross-sectional Graph Attention Network branch — here a directed volume→Parkinson weighted graph — change the forecast relative to the same-feature no-graph LSTM? To answer these we use a *masked panel*: rather than intersecting to the dates on which every ticker is present, we keep the union of trading dates with node and target masks for tickers not yet listed on a given date, so late-listed tickers are included without dropping whole dates and every model — including the graph model — is trained and evaluated on the same panel with the same five features.

Our contributions are: (i) a multi-horizon benchmark on the masked VN30/VN100 panel that holds the feature set and the data design fixed across HAR, an LSTM and an LSTM+GAT, so any difference is attributable to a single component; (ii) per-horizon results on all five metrics with the date-clustered DM test — HAR has the lowest QLIKE at every horizon with no learned model significantly lower, the deep models have the lowest short-horizon MAE on VN100, and the LSTM+GAT has a significantly lower QLIKE than the no-graph LSTM at $h1$ and $h5$ in both panels but not at $h10$ or $h22$; and (iii) a methodological contribution — the date-clustered DM test, since a naive per-observation test overstates significance by roughly the square root of the number of stocks. Every number is taken from a stored results file.

## 2 Related Work

**HAR and classical models.** HAR [1] approximates the long memory of realized volatility with a daily/weekly/monthly cascade; recent work reports that rolling-window and specification choices, rather than model class, often account for measured differences from HAR [12]. HARQ [8] exploits measurement error, and GARCH-family models remain standard conditional-variance benchmarks. We use HAR as the linear baseline and GARCH(1,1) as a classical benchmark. **Deep and graph models.** LSTMs [3] are widely applied to financial series, and graph neural networks model cross-asset spillovers, e.g. Graph Attention Networks [4], multivariate realized-volatility GNNs [9] and spillover-aware GNN-HAR hybrids [10]; reported effects vary across studies and data designs. Volatility spillover measurement has a long econometric tradition [11]. **Graph construction.** Our graph uses a directed volume→Parkinson relation (each node attends to its Top-5 volume-shock leaders, edge-weighted, over two hops), motivated by volume leading volatility. **Evaluation.** Volatility is latent, so we use the proxy-robust QLIKE loss [7] alongside MSE, with significance by the Diebold–Mariano test [5] and the Harvey–Leybourne–Newbold small-sample correction [6].

## 3 Method

**Target and features.** The target is the Parkinson variance [2] at day $t+h$, a non-negative point forecast. The primary target is short-term, horizons $h \in \{1, 5\}$ (one trading day and one trading week ahead); $h \in \{10, 22\}$ (two weeks and roughly one trading month) are an extended study. All three feature-based models (HAR, LSTM, LSTM+GAT) use the same five node features per ticker at $t$: the daily Parkinson variance, its 5-day rolling mean (weekly), its 22-day rolling mean (monthly), a market Parkinson factor (the cross-sectional average Parkinson variance) and a 20-day volume z-score.

**Models.** *HAR* (baseline): pooled OLS on the five features, linear. *LSTM*: a two-layer LSTM (hidden 64) over the lookback window of the five features; a single pooled model is trained over all tickers, each standardized by its own scaler fit on training rows only, with a linear output inverse-transformed for evaluation (dropout, weight decay, gradient clipping, ReduceLROnPlateau, early stopping). *LSTM+GAT* (Fig. 1): the LSTM branch plus a Graph Attention Network branch [4] that reads the five node features at $t$ and attends over a directed volume→Parkinson Top-5, edge-weighted, two-hop graph estimated on training rows only; the branches are concatenated into an MLP head. The leave-one-out comparison against the same-feature LSTM isolates the graph's marginal contribution. *GARCH*: a GARCH(1,1) conditional-variance model fit per ticker on the training Parkinson-variance series (via pseudo-returns), a classical benchmark that forecasts the variance series directly and does not use the node features.

**Masked panel.** A GAT operates over a common-date cross-section, which naively forces a common-date intersection (only dates on which every ticker is present) that discards most ticker-days and places the whole test set in one recent period. We instead use a masked panel: the GAT operates over a common-date cross-section with node and target masks for tickers not yet listed on a given date, so late-listed tickers are included without dropping whole dates, and a mask-aware loss ignores absent tickers. Every model is trained and evaluated on the same panel with the same five features and split (VN100: 46,308 masked obs over 454 test dates at $h1$; VN30: 10,106 obs over 326 test dates at $h1$).

**Protocol.** Each configuration runs five seeds, up to 20 epochs with early stopping. We report MSE, RMSE, MAE, QLIKE and $R^2$, seed-averaged, with equal weight; predictions are floored at $10^{-2}$ times the per-node training mean, applied identically to every model. Significance uses the *date-clustered* Diebold–Mariano test: the per-observation loss differential is collapsed to one value per calendar date before the statistic (HLN correction, HAC lag $h-1$) is computed. This is the panel-correct treatment for a cross-section that shares each trading date; a naive per-observation test treats the effective sample as the number of tickers times the number of dates and inflates the statistic by roughly the square root of the number of stocks (about six on VN30, ten on VN100). We report two targeted contrasts — LSTM vs HAR and LSTM+GAT vs LSTM — on QLIKE and MAE.

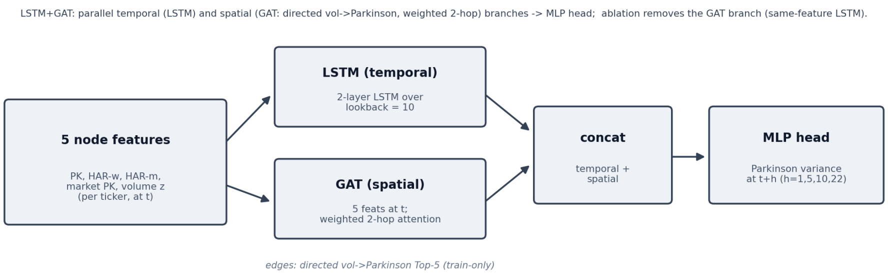


**Fig. 1.** LSTM+GAT: a per-node LSTM temporal branch and a GAT spatial branch (directed volume→Parkinson Top-5 weighted two-hop attention) are concatenated into an MLP head; the leave-one-out comparison removes the GAT branch to give the same-feature LSTM.

## 4 Data

Two panels are used with the Parkinson target: **VN30** (primary) and **VN100** (vnstock). Both use the masked panel of Sec. 3 with the five features, a single chronological split and per-ticker standardization fit on training rows only.

## 5 Experiments

We compare HAR, the five-feature LSTM and the five-feature LSTM+GAT (directed volume→Parkinson weighted two-hop) on VN30 and VN100 over the masked panel at horizons $\{1, 5, 10, 22\}$, reporting all five metrics and the two targeted date-clustered DM contrasts (LSTM vs HAR, LSTM+GAT vs LSTM).

## 6 Results

Point-error metrics are scaled (MSE $\times 10^{-7}$, RMSE $\times 10^{-4}$, MAE $\times 10^{-4}$); QLIKE and $R^2$ unscaled. All values are five-seed test means (`results/masked_rich_floor1e2/`). Row order is HAR → LSTM → LSTM+GAT. Tables 1 and 2 report all five metrics; Table 3 reports the two targeted date-clustered DM contrasts on QLIKE and MAE. The primary horizons $h1/h5$ appear first; the extended horizons $h10/h22$ appear in the lower block of each table.

**Short-horizon results ($h1$, $h5$).** At VN100 $h1$ the LSTM+GAT has the lowest value on all five metrics (MSE 2.362, RMSE 4.860, MAE 2.819, QLIKE 0.5107, $R^2$ 0.2251); by DM the LSTM has a lower MAE than HAR ($p < 0.001$), a higher QLIKE than HAR ($p = 0.028$), and the LSTM+GAT has a lower QLIKE than the no-graph LSTM ($p < 0.001$). At VN100 $h5$ HAR has the lowest MSE, RMSE, QLIKE and highest $R^2$ and the LSTM the lowest MAE (3.090); by DM the LSTM has a lower MAE than HAR ($p < 0.001$) and the LSTM+GAT a

**Table 1.** VN100 (46,308 masked obs over 454 test dates at $h1$; 5-seed means). Lower is better for MSE/RMSE/MAE/QLIKE, higher for $R^2$. † marks metrics where LSTM+GAT improves over the no-graph LSTM. MSE $\times 10^{-7}$, RMSE $\times 10^{-4}$, MAE $\times 10^{-4}$.

| $h$ | Model | MSE | RMSE | MAE | QLIKE | $R^2$ |
|---|---|---|---|---|---|---|
| 1 | HAR | 2.367 | 4.865 | 2.898 | 0.5115 | 0.2236 |
| 1 | GARCH | 5.917 | 7.692 | 5.195 | 0.7964 | -0.9407 |
| 1 | LSTM | 2.370 | 4.869 | 2.821 | 0.5525 | 0.2224 |
| 1 | LSTM+GAT | **2.362**† | **4.860**† | **2.819**† | **0.5107**† | **0.2251**† |
| 5 | HAR | **2.606** | **5.104** | 3.160 | **0.5633** | **0.1466** |
| 5 | GARCH | 5.843 | 7.644 | 5.197 | 0.7865 | -0.9136 |
| 5 | LSTM | 2.638 | 5.136 | **3.090** | 0.5841 | 0.1361 |
| 5 | LSTM+GAT | 2.635† | 5.133† | 3.103 | 0.5690† | 0.1371† |
| *Extended horizons* | | | | | | |
| 10 | HAR | **2.754** | **5.248** | 3.306 | **0.6023** | **0.0978** |
| 10 | GARCH | 6.399 | 7.999 | 5.306 | 0.8018 | -1.0957 |
| 10 | LSTM | 2.798 | 5.289 | 3.282 | 0.6070 | 0.0837 |
| 10 | LSTM+GAT | 2.802 | 5.294 | **3.276**† | 0.6072 | 0.0822 |
| 22 | HAR | **2.891** | **5.377** | 3.486 | **0.6405** | **0.0544** |
| 22 | GARCH | 7.089 | 8.420 | 5.360 | 0.7928 | -1.3184 |
| 22 | LSTM | 2.979 | 5.458 | **3.468** | 0.6518 | 0.0257 |
| 22 | LSTM+GAT | 3.003 | 5.480 | 3.563 | 0.6544 | 0.0178 |

lower QLIKE than the LSTM ($p = 0.022$). At VN30 $h1$ the LSTM+GAT has the lowest MSE, RMSE, MAE and highest $R^2$ and HAR the lowest QLIKE (0.5159); by DM the LSTM has a higher QLIKE than HAR ($p < 0.001$) and the LSTM+GAT a lower QLIKE ($p < 0.001$) and squared error ($p = 0.003$) than the no-graph LSTM. At VN30 $h5$ HAR has the lowest value on all five metrics; by DM the LSTM has a higher QLIKE ($p = 0.037$) and MAE ($p < 0.001$) than HAR and the LSTM+GAT a lower QLIKE than the LSTM ($p = 0.031$). Across the four short-horizon cells no learned model has a significantly lower QLIKE than HAR, and the LSTM+GAT has a significantly lower QLIKE than the no-graph LSTM in all four, equal to HAR's QLIKE level at VN100 $h5$ (0.5690 vs 0.5633) and $h1$ (0.5107 vs 0.5115).

**Extended-horizon results ($h10$, $h22$).** HAR has the lowest MSE, RMSE, QLIKE and highest $R^2$ on both panels at both horizons; the LSTM has the lowest MAE at VN100 $h22$ (3.468) and VN30 $h10$ (2.721), the LSTM+GAT the lowest MAE at VN100 $h10$ (3.276). By DM the LSTM+GAT-vs-LSTM QLIKE difference is not significant at $h10$ or $h22$ on either panel ($p \geq 0.55$); the LSTM has a higher squared error than HAR at VN100 $h10$ ($p = 0.022$) and $h22$ ($p = 0.002$) and a higher MAE than HAR at VN30 $h22$ ($p = 0.001$).

**GARCH benchmark.** GARCH(1,1) has a higher MSE, RMSE, MAE and QLIKE and a lower $R^2$ than HAR at every horizon on both panels, with a

**Table 2.** VN30 (10,106 masked obs over 326 test dates at $h1$; 5-seed means). $^{\dagger}$ as in Table 1. Same scaling as Table 1.

| $h$ | Model | MSE | RMSE | MAE | QLIKE | $R^2$ |
|---|---|---|---|---|---|---|
| 1 | HAR | 1.927 | 4.389 | 2.389 | **0.5159** | 0.2308 |
| 1 | GARCH | 2.770 | 5.263 | 3.453 | 0.7911 | -0.1060 |
| 1 | LSTM | 1.929 | 4.393 | 2.407 | 0.6073 | 0.2297 |
| 1 | LSTM+GAT | **1.912**$^{\dagger}$ | **4.372**$^{\dagger}$ | **2.366**$^{\dagger}$ | 0.5800$^{\dagger}$ | **0.2368**$^{\dagger}$ |
| 5 | HAR | **2.139** | **4.625** | **2.583** | **0.5965** | **0.1497** |
| 5 | GARCH | 2.785 | 5.277 | 3.492 | 0.7864 | -0.1071 |
| 5 | LSTM | 2.164 | 4.652 | 2.632 | 0.6402 | 0.1397 |
| 5 | LSTM+GAT | 2.147$^{\dagger}$ | 4.633$^{\dagger}$ | 2.651 | 0.6059$^{\dagger}$ | 0.1467$^{\dagger}$ |
| *Extended horizons* | | | | | | |
| 10 | HAR | **2.301** | **4.797** | 2.733 | **0.6428** | **0.1028** |
| 10 | GARCH | 2.835 | 5.324 | 3.503 | 0.7791 | -0.1052 |
| 10 | LSTM | 2.310 | 4.806 | **2.721** | 0.6584 | 0.0995 |
| 10 | LSTM+GAT | 2.326 | 4.823 | 2.725 | 0.6564$^{\dagger}$ | 0.0931 |
| 22 | HAR | **2.272** | **4.766** | **2.785** | **0.6422** | **0.0723** |
| 22 | GARCH | 2.709 | 5.205 | 3.413 | 0.7445 | -0.1064 |
| 22 | LSTM | 2.383 | 4.881 | 2.904 | 0.6550 | 0.0271 |
| 22 | LSTM+GAT | 2.385 | 4.883 | 2.884$^{\dagger}$ | 0.6548$^{\dagger}$ | 0.0261 |

negative $R^2$ on both; its QLIKE is higher than HAR's under the date-clustered DM test in all eight cells ($p < 0.001$, except VN30 $h22$ where $p = 0.001$).

**Per-metric ranking.** No single metric determines the ordering, so all five are reported with equal weight: HAR has the lowest QLIKE at every horizon; the deep and graph models have the lowest point error at the short horizons; HAR has the lowest MSE/RMSE from $h5$ onward.

**Cross-market check (S&P 500, exploratory).** As a preliminary out-of-sample check on a larger market, the same masked panel and five features were applied to S&P 500 constituents (442 nodes, 1,624 test dates) with a single seed; Table 4 reports all five metrics. In contrast to the Vietnamese panels, the deep models attain the lowest point error (MSE, RMSE, MAE and $R^2$) from $h5$ onward. HAR retains the lowest QLIKE at $h5$, $h10$ and $h22$; at $h1$ the LSTM+GAT has a marginally lower QLIKE. Two caveats apply: the run uses a single seed, and the deep models' QLIKE at $h5$–$h22$ is inflated by the per-node positivity floor at the S&P 500 variance scale, so QLIKE for the deep models on this panel is not yet reliable and a scale-aware floor with multiple seeds is left for future work.

**Output parameterization and QLIKE stability (robustness).** The deep model's QLIKE was sensitive to the random seed under the additive standardized-target design with a post-hoc floor. Replacing that target with a node-scaled ratio (the model predicts $y_t/\bar{y}_i$ and reconstructs the forecast as the ratio times the training-period node mean) reduces the five-seed QLIKE standard deviation

**Table 3.** Targeted date-clustered DM contrasts ($p$-value; favoured model in parentheses; bold if $p < 0.05$). H=HAR, L=LSTM, G=LSTM+GAT, on QLIKE and MAE.

| | | LSTM vs HAR | | LSTM+GAT vs LSTM | |
|---|---|---|---|---|---|
| Panel | $h$ | QLIKE | MAE | QLIKE | MAE |
| VN100 | 1 | **.028 (H)** | **<.001 (L)** | **<.001 (G)** | .684 (G) |
| VN100 | 5 | .148 (H) | **<.001 (L)** | **.022 (G)** | **.002 (L)** |
| VN100 | 10 | .598 (H) | .201 (L) | .872 (L) | .064 (G) |
| VN100 | 22 | .429 (H) | .611 (L) | .658 (L) | **<.001 (L)** |
| VN30 | 1 | **<.001 (H)** | .221 (H) | **<.001 (G)** | **<.001 (G)** |
| VN30 | 5 | **.037 (H)** | **<.001 (H)** | **.031 (G)** | **.015 (L)** |
| VN30 | 10 | .260 (H) | .340 (L) | .547 (G) | .624 (L) |
| VN30 | 22 | .318 (H) | **.001 (H)** | .936 (G) | **.021 (G)** |

from 0.043 to 0.005 (VN30) and from 0.039 to 0.004 (VN100), with the same linear output and floor, identifying node scaling rather than the positive output link as the primary source of stability. Positive links (exponential or softplus) provide smaller additional improvements and remove the post-hoc floor; no statistically significant QLIKE difference was detected between them at the 5% level after bias-matched initialization. HAR retains a lower mean QLIKE, but its advantage over the ratio-parameterized deep model is not statistically significant under the date-clustered DM test (VN30 $p = 0.29$, VN100 $p = 0.60$); a failure to reject does not establish equivalence. This indicates the earlier instability was largely attributable to the target parameterization rather than the forecasting architecture. Table 5 reports the five-seed QLIKE.

## 7 Discussion

On both panels HAR has the lowest QLIKE at every horizon, and no learned model has a significantly lower QLIKE than HAR under the date-clustered DM test. At $h1$ and $h5$ the LSTM and LSTM+GAT have the lowest MAE on VN100 (LSTM vs HAR, $p < 0.001$), and the LSTM+GAT has a significantly lower QLIKE and, on VN30, lower squared error than the same-feature no-graph LSTM; at VN100 $h1$ and VN30 $h1$ the LSTM+GAT has the lowest value on four or five of the metrics. At $h10$ and $h22$ HAR has the lowest MSE, RMSE, QLIKE and $R^2$, and the LSTM+GAT-vs-LSTM QLIKE difference is not significant. The graph's effect on QLIKE relative to the no-graph LSTM is present at the short horizons and absent at the longer horizons; the deep models' MAE reduction is present at the short horizons on VN100 and reverses at the longer horizons, where their squared error and MAE are higher than HAR's. The ranking is metric- and horizon-dependent, which is why all five metrics are reported with equal weight. The date-clustered DM test matters for these conclusions: a naive per-observation test on this cross-sectionally dependent panel inflates the

**Table 4.** S&P 500 (exploratory, single seed): the same masked panel and five features as Tables 1–2, 442 nodes, 1,624 test dates. Lower is better for MSE/RMSE/MAE/QLIKE, higher for $R^2$; bold = best per horizon. HAR is the same five-feature model as in Tables 1–2. GARCH not run on this panel.

| $h$ | Model | MSE | RMSE | MAE | QLIKE | $R^2$ |
|---|---|---|---|---|---|---|
| 1 | HAR | **6.171** | **7.856** | **2.361** | 0.4078 | **0.3195** |
| | LSTM | 6.367 | 7.979 | 2.453 | 0.4110 | 0.2979 |
| | LSTM+GAT | 6.384 | 7.990 | 2.466 | **0.4064** | 0.2960 |
| 5 | HAR | 8.062 | 8.979 | 2.691 | **0.4303** | 0.1110 |
| | LSTM | **7.850** | **8.860** | **2.573** | 1.0433 | **0.1344** |
| | LSTM+GAT | 7.902 | 8.890 | 2.637 | 0.7907 | 0.1287 |
| 10 | HAR | 8.948 | 9.459 | 2.842 | **0.4775** | 0.0153 |
| | LSTM | 8.505 | 9.222 | 2.767 | 0.8276 | 0.0640 |
| | LSTM+GAT | **8.497** | **9.218** | **2.701** | 1.0110 | **0.0650** |
| 22 | HAR | 9.623 | 9.810 | 2.983 | **0.6150** | -0.0576 |
| | LSTM | 9.399 | 9.695 | 3.259 | 0.8515 | -0.0330 |
| | LSTM+GAT | **9.172** | **9.577** | **2.952** | 1.3198 | **-0.0080** |

**Table 5.** Output-parameterization robustness (LSTM, $h5$, 5 seeds): QLIKE mean±std. A: z-score + linear + floor (original); B: ratio + linear + floor; C: ratio + exp (no floor); D: ratio + softplus (no floor). HAR QLIKE is 0.5962 (VN30) and 0.5694 (VN100).

| Panel | A | B | C | D |
|---|---|---|---|---|
| VN30 | $0.660 \pm 0.043$ | $0.613 \pm 0.005$ | $0.610 \pm 0.002$ | $0.610 \pm 0.003$ |
| VN100 | $0.602 \pm 0.039$ | $0.572 \pm 0.004$ | $0.573 \pm 0.003$ | $0.576 \pm 0.007$ |

statistic by roughly the square root of the number of stocks, so a difference that is not significant per date can appear significant per observation.

## 8 Conclusion

For daily Parkinson-variance forecasting on the Vietnamese market, evaluated with the same features, the same masked panel and the same date-clustered Diebold–Mariano inference, HAR has the lowest QLIKE at every horizon on both panels and no learned model has a significantly lower QLIKE than HAR. At the short horizons ($h1$, $h5$) the LSTM and LSTM+GAT have the lowest MAE on VN100, and the directed volume→Parkinson weighted graph gives the LSTM+GAT a significantly lower QLIKE than the same-feature no-graph LSTM in both panels, equal to HAR's QLIKE level at VN100 $h5$; at the extended horizons ($h10$, $h22$) HAR has the lowest MSE, RMSE, QLIKE and $R^2$, and the graph's QLIKE difference from the no-graph LSTM is not significant. The ranking is metric- and horizon-dependent, so all five metrics are reported with equal weight.

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